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**Roll No.** | | | | | | | | | | | | | **Total No. of Pages : 02**

**Total No. of Questions : 09**

**B.Tech. (CE) (Sem.–6)**

# DESIGN OF STEEL STRUCTURES

**Subject Code : PECE-603B-18**

**M.Code : 79401**

**Date of Examination : 19-12-2022**

**Time : 3 Hrs.** **Max. Marks : 60**

**INSTRUCTIONS TO CANDIDATES :**

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

## SECTION-A

**1. Write briefly :**

a) What are the elements of plate girder?

b) What are different types of beam columns connections?

c) Write a note on slab base.

d) What is column bracket?

e) Write a short note on Plastic hinge mechanism.

f) Explain the main function of bearings.

g) Differentiate between lateral load and longitudinal on the gantry girder.

h) What are various types of stiffeners?

i) Write down types of Industrial buildings.

j) Write a note on the basis for eccentrically loaded columns.

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## SECTION-B

2. a) What are the assumptions made in plastic analysis of beams?

   b) A fixed beam of span L carries a UDL (W) on the left half portion using plastic theory determine the value of W at collapse. The plastic moment of resistance of beam is Mp.

3. Write a detailed note on Grillage foundation. Explain with the help of a diagram.

4. Write a detailed note on seat and framed connections.

5. Write a detailed note on: a) Elements of plate girder b) Steps involved in the design of plate girder.

6. What are the different types of bracings used in a braced building?

## SECTION-C

7. Design a gantry girder to carry an electric overhead travelling crane for the following data :

   Crane Capacity = 260 kN

   Self weight of crane alone = 200 kN

   Self weight of trolley, electric motor, hook etc = 70 kN

   Minimum approach of crane hook = 1.2 m

   Distance between centers of crane wheels = 3.5 m

   Distance between cranes of gantry girders = 14 m

   Span of gantry girder = 8 m

   Weight of rail section = 0.300kN per metre.

8. A plate girder with Fe 450 steel plates is having 12mm × 1800 mm web plates and 75mm × 550mm flange plates. Determine the design flexural strength if Un-restrained length is 10m.

9. a) i) Write a short note about the framed connections and their design.

      ii) Discuss the various functions of bearing.

   b) ISMB 350@ 524N/m is to join a column ISHB 300@588N/m. The beam has to transmit end reaction of 230kN. Design a stiffened seat connection.

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